

Automatic Fall-Detection Pendant

Strengths of the Proposal

The pendant responds to a clearly articulated safety need: an older adult who falls may be unconscious, disoriented, physically unable to reach a phone, or unfamiliar with smartphone interfaces. Automatic fall detection combined with a 60-second cancellation period is a sensible human-in-the-loop design because it permits intervention while reducing unnecessary alerts. The pendant form is familiar, portable, and more accessible than a complex mobile application. The proposed materials are affordable, making a functional prototype achievable within the development period.

Direct Critical Assessment

The proposal treats “detect motion” and “detect a dangerous fall” as if they were the same problem. They are not. Sitting abruptly, dropping the pendant, lying down, bending, or exercising can all produce fall-like signals, while a slow collapse may produce no obvious impact. The proposed 60-second cancellation timer does not solve false detection if the pendant itself is dropped in another room, and it may delay help when a genuine user is unconscious. The diagram also jumps directly from sensing to contacting family or paramedics without specifying Bluetooth, Wi-Fi, cellular service, a paired phone, subscription costs, indoor coverage, or what happens during network failure. A device advertised for emergencies but lacking a manual SOS control, battery warning, waterproofing, and staged escalation is not yet dependable. Random Forest, Extra Trees, or XGBoost could classify engineered accelerometer features in a prototype, but a small dataset of classmates pretending to fall would not prove reliability for elderly users with different movement patterns.

Recommended Next Step

Create a prototype using an accelerometer-enabled microcontroller and test it against a labelled dataset of falls and ordinary activities. Report sensitivity, false-alarm rate, alert latency, and battery life. Use a staged workflow: audible warning, cancel button, caregiver notification, then emergency escalation according to an approved protocol.